Electrical and Computer Engineering Course Syllabus

010153002 Computer Programming

Semester 1 / Academic Year 2026

Department of Electrical and Computer Engineering (ECE)
Faculty of Engineering, King Mongkut's University of Technology North Bangkok

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1. Course Information

Course Number:	010153002
Course Name:	Computer Programming
Credits:	3 (2–2–5)
Contact Hours:	4 hours per week (2 lecture + 2 laboratory)
Schedule:	Every Thursday, 13:00–17:00
Semester Dates:	22 June 2026 – 11 October 2026
Course Type:	Required
Prerequisites:	None
Language:	English (supplementary explanations in Thai available)
Textbook:	Kernighan, B. W. & Ritchie, D. M. *The C Programming Language*, 2nd Edition. Prentice Hall, 1988.

2. Course Description

Basic operations of a computer, required components of a computer, how hardware and software work together, data processing, fundamental of high level computer programming, application design and development, problem solving by using computer programming.

3. Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

1. Understand computer systems concepts, different computing environments, computer languages, and the system development life cycle.
2. Write programs using basic programming constructs including identifiers, data types, variables, constants, input/output statements, logical and comparative operators, and control structures (selections and loops).
3. Understand and apply expressions, precedence rules, associativity in evaluating expressions, type conversions, and various statement types.
4. Develop and implement programs using arrays, pointers, and functions, including sorting algorithms, search algorithms, and string manipulations.
5. Understand and utilize complex data structures such as structures, unions, and enumerated types.
6. Recognize the importance of effective communication, ethical and professional responsibilities in engineering, and the impact of engineering solutions in various contexts.

4. ABET Student Outcomes Addressed

ABET Student Outcome (SO)	Course Learning Outcome (CLO)
1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1–6
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	1–6
3. An ability to communicate effectively with a range of audiences.	6
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	6
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	—
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	1–6
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	1–6

5. CLO Assessment Mapping

Course Learning Outcome (CLO)	SO	Assessment Tools
1. Understand computer systems concepts, different computing environments, computer languages, and the system development life cycle.	1, 7	Exams, Quiz, Assignment, Project
2. Write programs using basic programming constructs including identifiers, data types, variables, constants, input/output statements, logical and comparative operators, and control structures (selections and loops).	1, 2, 7	Exams, Quiz, Assignment, Project
3. Understand and apply expressions, precedence rules, associativity in evaluating expressions, type conversions, and various statement types.	1, 2, 7	Exams, Quiz, Assignment, Project
4. Develop and implement programs using arrays, pointers, and functions, including sorting algorithms, search algorithms, and string manipulations.	1, 2, 7	Exams, Quiz, Assignment, Project
5. Understand and utilize complex data structures such as structures, unions, and enumerated types.	1, 7	Exams, Quiz, Assignment, Project
6. Recognize the importance of effective communication, ethical and professional responsibilities in engineering, and the impact of engineering solutions in various contexts.	1, 7	Exams, Quiz, Assignment, Project

6. Weekly Schedule

All classes meet on **Thursday, 13:00–17:00**. Lecture: 13:00–15:00 | Laboratory: 15:00–17:00.

Week	Date	Lecture (13:00–15:00)	Lab (15:00–17:00)
1	Thu 25 Jun 2026	**Lec 1:** Introduction to Computer & Programming Computer components, hardware/software interaction, compilation pipeline, development environment.	**Lab 1:** Introduction to C Compile and run `hello_world.c`; explore compiler flags.
2	Thu 02 Jul 2026	**Lec 2:** Structure of a Program & Data Types Identifiers, variables, constants; integer, float, char; `printf`/`scanf`; type conversions.	**Lab 2:** Expressions & Operators Arithmetic operators, precedence rules, pre/post increment.
3	Thu 09 Jul 2026	**Lec 3:** Expressions & Operators Arithmetic, logical, and comparative operators; associativity; precedence; various statement types.	**Lab 3:** Expressions Practice Type conversion exercises; operator precedence tasks.
4	Thu 16 Jul 2026	**Lec 4:** Selection – Making Decisions `if`, `if-else`, nested `if`, `switch`; ternary operator.	**Lab 4:** Conditional Statements Grade classifier, switch on user input.
5	Thu 23 Jul 2026	**Lec 5:** Repetition `while`, `do-while`, `for`; `break`, `continue`; nested loops.	**Lab 5:** Loop Control Summation, factorial, pattern printing, input validation.
6	Thu 30 Jul 2026	**Lec 6:** Functions & Macros Declaration, definition, call, return values; scope; recursion; `#define` macros.	**Lab 6:** Functions Write and call helper functions; implement recursive factorial.
7	Thu 06 Aug 2026	**Lec 7:** Arrays 1-D and 2-D arrays; traversal; passing arrays to functions; sorting and searching algorithms.	**Lab 7:** Arrays Statistics on scores array; bubble sort; binary search.
8	Thu 13 Aug 2026	Part 1 Review & Consolidation Q&A on Lectures 1–7; past exam walkthrough; common pitfalls.	**Lab 8:** Pointers (Part A) Address-of & dereference; pointer tracing; swap via pointers. *(Lec 8 content distributed as pre-reading.)*
—	**Thu 20 Aug 2026**	**University Midterm Examination Period (17–23 August 2026)** **No class this week.** Midterm exam date & venue announced by the Registrar. Covers: Lectures 1–7, Labs 1–7.	
9	Thu 27 Aug 2026	**Lec 8:** Pointers Address-of & dereference; pointer arithmetic; `malloc`/`free`; dynamic memory.	**Lab 8:** Pointers (continued) Dynamic array allocation; pointer to function.

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Week	Date	Lecture (13:00–15:00)	Lab (15:00–17:00)
10	Thu 03 Sep 2026	**Lec 9:** Strings Null-terminated `char` arrays; `strlen`, `strcpy`, `strcmp`, `strcat`; pointer traversal; string manipulation.	**Lab 9:** Strings `my_strlen`; reverse in place; palindrome checker; count vowels. *Project assignment announced.*
11	Thu 10 Sep 2026	**Lec 10:** Enumerated, Structure, and Union Types `struct` definition; `.` vs `->`; pass-by-pointer; `typedef`; `union`; `enum`; memory padding.	**Lab 10:** Structures & Unions Student record with `->`; sort structs; tagged union area calculator.
12	Thu 17 Sep 2026	Project Work Session Instructor and TA available for project Q&A; individual code review on request.	Course Review & Exam Preparation Past exam questions; common bugs; tracing exercises.
13	Thu 24 Sep 2026	Course Wrap-up & Q&A Open Q&A on all topics; exam strategy; feedback session. **Last class.** *Project submission deadline: 17:00 today.*	
—	**TBD**	**Final Examination** *Date & venue announced by the Registrar* Covers **Part 2 only**: Lectures 8–10 (Pointers, Strings, Structs/Unions) and Labs 8–10. Closed-book; no computer; no AI tools.	

Note: Teaching concludes on **24 September 2026** (Week 13). The midterm exam period (17–23 Aug) and final exam dates are set by the Registrar. The **final exam covers Part 2 only** (Lectures 8–10).

7. Assessment and Grading

Assessment Breakdown

- Lab Exercises, Quizzes & Assignment 40%
- Midterm Examination 30%
- Final Examination 30%

A minimum attendance of 80 % (11 of 13 sessions) is required to be eligible to sit the Final Examination.

Grading Scale

80–100	A	60–64	C
75–79	B+	55–59	D+
70–74	B	50–54	D
65–69	C+	0–49	F

8. Topics Covered

a. Introduction to Computer
b. Introduction to Computer Language
c. Structure of a Program
d. Selection – Making Decisions
e. Repetition
f. Arrays
g. Pointers
h. Functions
i. Strings
j. Enumerated, Structure, and Union Types

9. Course Policies

Examinations. Both examinations are **closed-book and paper-based** (no computer, no AI tools). Students must bring their student ID card. Questions include multiple-choice, code tracing, bug fixing, and short programming tasks written by hand. No make-up exams except with documented medical or official university reasons approved *before* the exam date wherever possible.

Academic Integrity. Students are expected to submit their own work. The use of AI tools to generate code submitted as individual work is not permitted. Plagiarism and academic dishonesty are subject to university disciplinary procedures which may include a grade of F for the course.

Lab Work. Lab sheets must be submitted by the end of the lab session. Late submissions are not accepted without a valid excuse approved in advance.

Project. Individual assignment; collaboration on code is not permitted. Submitted code must compile and run on GCC without modification. Include a short written report (one page) describing your design choices.

Communication. Primary channel is the course LMS (announced in Week 1). Email responses within two working days.

10. Course Materials

- Lecture slides: ruslylove.github.io/compro
- Lab sheets: available on the course LMS before each class
- Compiler: GCC (Linux/macOS) or MinGW-GCC (Windows); any IDE is acceptable
- *Reference:* Kernighan & Ritchie, *The C Programming Language*, 2nd ed. (Prentice Hall, 1988)

This syllabus is subject to minor adjustments at the instructor's discretion to accommodate the academic calendar and class progress. Students are responsible for staying informed of any changes announced in class or via the course LMS.